21-8 **Graphs of Parabolas**

The graph of a first-degree polynomial function (a linear function) is a ____________________________ .
For example $y = 2x + 5$ is a straight line.

The graph of a **second-degree polynomial function** (____________________________________),
is not a straight line and requires a larger number of points to draw its graph.

The graph of a quadratic function of the form $\mathbf{y = ax^2 + bx + c}$,
where a, b and c are real numbers and $a \neq 0$, is a ___________________________.

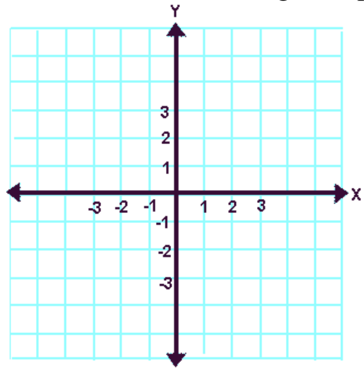
An easy method for graphing parabolas involves preparing a chart.
Of course, the graphing calculator can also be used.

Example:

Graph the parabola $y = x^2 - 4x$
on the interval from $x = -1$ to $x = 5$.

x	$x^2 - 4x$	y
-1		
0		
1		
2		
3		
4		
5		

Plot the points generated in the table.
Draw a smooth curve through the points.

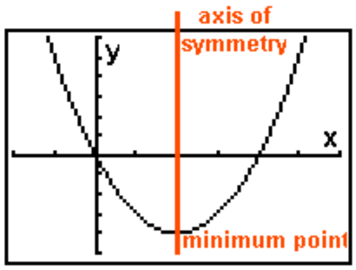


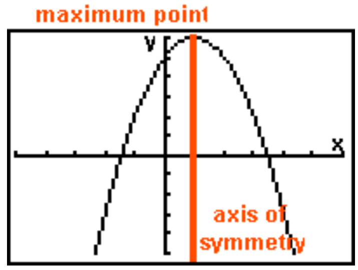
The points where the graph crosses the x-axis
are called the __________________.
The parabola crosses the x-axis at ________ and ________.

The _________________________________is a vertical line passing through the turning point of a parabola.

In this example the turning point is ________.   The equation of the axis of symmetry is ____________.

Parabolas are of the form: $y = ax^2 + bx + c$





If **a** is_______________, the parabola opens
upward and has a minimum point.
The axis of symmetry is **$x = (-b)/2a$**

If **a** is _____________, the parabola opens
downward and has a maximum point.
The axis of symmetry is **$x = (-b)/2a$.**

Using the Graphing Calculator to Investigate Parabolas

Lab Sheet - Investigating Parabolas

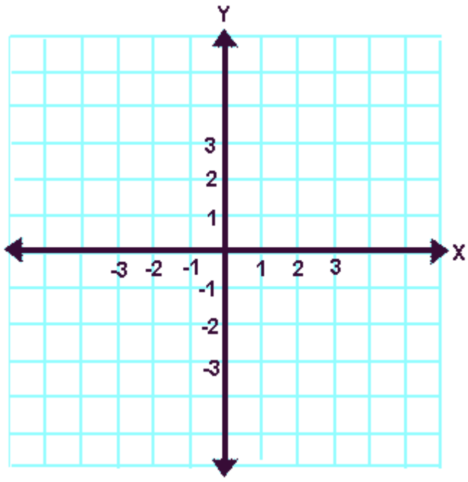
1) Graph the following equations using the graphing calculator.

2) Use the STANDARD window (ZOOM 6).

3) Answer the question associated with each problem.

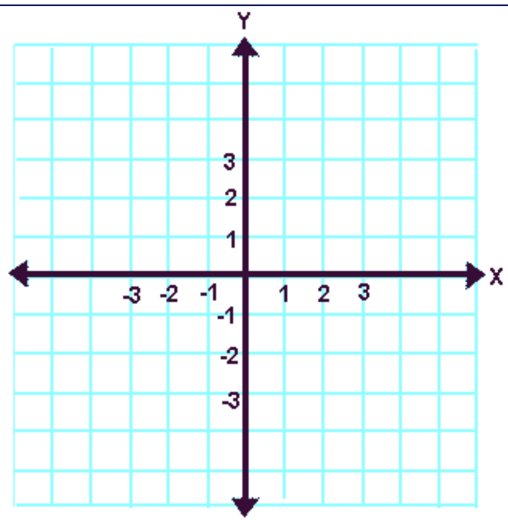
1. **$Y1= x^2$**
   $Y2= x^2 + 2$
   $Y3= x^2 + 4$

   What happens to the graph when a number is added to x^2?



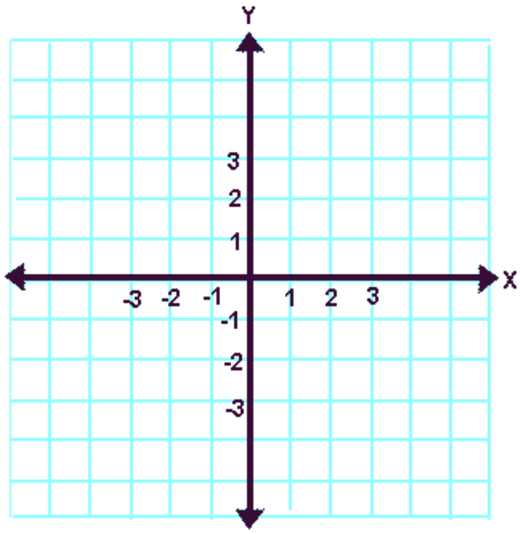
2. **$Y1= x^2$**
   $Y2= x^2 - 5$
   $Y3= x^2 - 2.5$

   What happens to the graph when a number is subtracted from x^2?



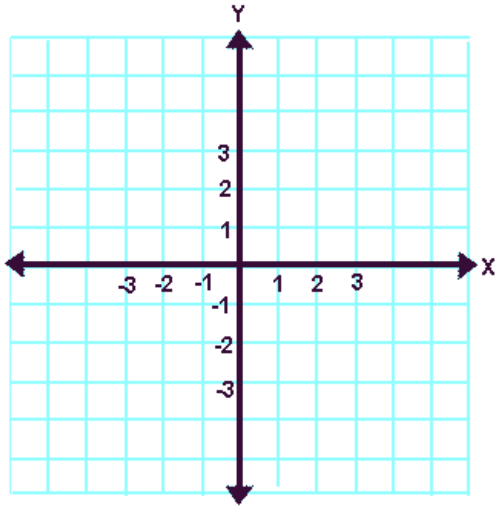
3. **$Y1= x^2$**
   $Y2= 2x^2$
   $Y3= 6x^2$

   What happens to the graph when x^2 is multiplied by a number greater than 1?



4. **Y1= x^2**
Y2= $.5x^2$
Y3= $.2x^2$

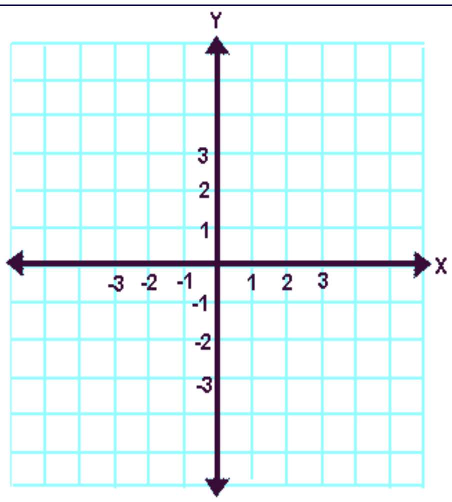
What happens to the graph when x^2 is multiplied by a number between 0 and 1?



5. **Y1= x^2**
Y2= $-x^2$
Y3= $-2x^2$

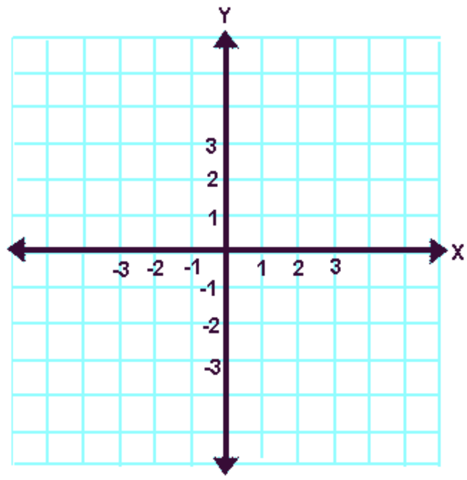
be careful to use the
egation key and not the
ubtraction key)

What happens to the graph when the coefficient of x^2 is negative?



6.
Y1= $2x^2 + 3$
Y2= $-2x^2 + 3$
Y3 = x^2

Compare these two graphs with the third. What observations can be made?



7.
Y1= $.5x^2 - 2$
Y2= $-.5x^2 + 2$

Compare these two graphs. What observations can be made?